Positive, monotone, and convex Gaussian process regression with Laplacian eigenfunctions

Supplementary material

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A Proofs and supporting results

A.1 Gaussian conditioning and its specializations

Theorem 1 (Gaussian conditioning under a continuum sign restriction)**.** *Under the finite Gaussian model in Section 2, let* $\boldsymbol{a} = (a_0, \dots, a_m)^T$ *denote the coefficient vector and fix* $r \in \{0, 1, 2\}$*. Suppose an invertible linear change of coordinates* $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ *gives* $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h_r$*. Here* h_r *is a fixed function in the retained span, chosen to have a uniformly positive* r*th derivative on* I*, and* γ *is its scalar coefficient. The curve* $f_{\boldsymbol{g}}$ *depends linearly on the remaining coordinates* $\boldsymbol{g}$*. Assume these functions belong to* $C^r(I)$ *with* $\min_{x \in I} \frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x) > 0$*. In these coordinates, the ordinary Gaussian posterior has the form*

$$\boldsymbol{g} \mid \boldsymbol{y} \sim N(\boldsymbol{m}_g, \boldsymbol{V}_g), \qquad \gamma \mid \boldsymbol{g}, \boldsymbol{y} \sim N(\mu(\boldsymbol{g}), s^2). \tag{4}$$

Here $\boldsymbol{V}_g$ *is positive definite,* μ *is affine, and* $s > 0$ *is constant in* $\boldsymbol{g}$*. The finite, attained threshold*

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \frac{-\frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x)} \tag{5}$$

satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_r$ *exactly when* $\gamma \geq \tau_r(\boldsymbol{g})$*. This event has positive ordinary posterior probability and a null boundary in coefficient space. Its marginal density for* $\boldsymbol{g}$ *with respect to Lebesgue measure on* $\mathbb{R}^m$ *is*

$$p(\boldsymbol{g} \mid \boldsymbol{y}, \mathcal{C}_r) = \frac{p(\boldsymbol{g} \mid \boldsymbol{y}) \Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)}{P(f_{\boldsymbol{a}} \in \mathcal{C}_r \mid \boldsymbol{y})}. \tag{6}$$

Given this draw of $\boldsymbol{g}$*, draw* γ *from the conditional normal in Equation* (4) *truncated below at* $\tau_r(\boldsymbol{g})$ *and apply the inverse coefficient map. Equivalently, for an independent* $U \sim \mathrm{Uniform}(0, 1)$*,*

$$\gamma = \mu(\boldsymbol{g}) - s\Phi^{-1}\big(U\Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)\big). \tag{7}$$

This reconstructs the full conditioned coefficient law, under which $\min_I \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r} > 0$ *almost surely.*

Proof of Theorem 1. Fix r, I, the invertible map, and the ordinary posterior under the finite Gaussian model in Section 2 and the positive-direction hypotheses of Theorem 1, and write $b_r = \min_{x \in I} \frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x) > 0$. The role of the positive direction is to turn every derivative inequality into a lower bound on the same scalar coefficient: division by $\frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}$ gives the threshold equivalence,

and continuity on compact I, together with $b_r > 0$, makes Equation (5) finite and attained. Since the invertible map preserves positive definiteness of the posterior covariance, Gaussian conditioning (Rasmussen and Williams, 2006, Section 2.2) gives Equation (4), including a positive Schur complement s^2.

Integrating the restricted joint Gaussian density over the scalar coefficient now leaves its conditional survival probability as a weight on $\boldsymbol{g}$, which proves Equation (6). This weight is strictly positive for every $\boldsymbol{g}$, so the conditioning event has positive probability. Given a draw from the weighted marginal, conditioning the Gaussian law of $\gamma \mid \boldsymbol{g}, \boldsymbol{y}$ on $\gamma \geq \tau_r(\boldsymbol{g})$ gives the stated truncated normal, whose survival-function inverse is Equation (7). Applying the inverse coefficient map therefore recovers the original coefficient law conditional on the sign restriction.

The minimum derivative is continuous in the coefficients, so coefficient vectors with a positive minimum lie in the event's interior and those with a negative minimum lie in its exterior. Conditional on $\boldsymbol{g}$, the nondegenerate normal assigns zero mass to equality at the threshold. At equality an attained maximizing location has zero derivative, whereas above it

$$\min_I \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r} \geq (\gamma - \tau_r(\boldsymbol{g}))b_r > 0.$$

At equality, an arbitrarily small increase or decrease in γ moves to either side of the event, identifying its coefficient boundary as precisely the threshold graph. Its zero Gaussian probability implies that the reconstructed law lies strictly above the graph almost surely, proving the strict-derivative assertion. □

Corollary 1 (Positivity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 1$ and $I \subset (c - L, c + L)$. Set $h_0 = 1$, $\boldsymbol{g} = (a_1, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = \sum_{j=1}^m a_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + a_0 h_0$, with inverse obtained by reinserting a_0.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define $P_{0,\boldsymbol{g}}(w) = \sum_{j=1}^m j a_j T_j(w)$. The threshold is

$$\tau_0(\boldsymbol{g}) = -\min_{x \in I} f_{\boldsymbol{g}}(x). \tag{8}$$

Theorem 1 applies with scalar coefficient $\gamma = a_0$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $P_{0,\boldsymbol{g}}$. This stationary polynomial has degree at most m when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the curve $f_{\boldsymbol{g}}$ is constant and either endpoint suffices.

Proof of Corollary 1. Use the finite Gaussian model in Section 2 and its compact physical interval. Separating the intercept is an invertible permutation of the coefficients, with $\min_I h_0 = 1$, so Theorem 1 at order zero gives Equation (8) and the conditioned law. To locate the minimum over the whole interval, use the Chebyshev identities to write

$$f_{\boldsymbol{g}}(x) = \frac{\sqrt{1 - w(x)^2}}{\sqrt{L}} \sum_{j=1}^m a_j U_{j-1}(w(x)), \qquad \frac{\mathrm{d}f_{\boldsymbol{g}}}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} P_{0,\boldsymbol{g}}(w(x)).$$

Because the cosine is strictly decreasing for $0 < \theta(x) < \pi$, interior stationary locations correspond exactly to interior roots of the displayed polynomial. The extreme value theorem and Fermat's theorem therefore give the endpoint and stationary candidates, with finitely many distinct roots, including repeated roots, when the polynomial is nonzero and has degree at most m. If it vanishes identically, the derivative is zero throughout I, so the mean value theorem gives constancy. Although the intercept is independent of the spectral coefficients in the prior, reconstruction uses its conditional posterior given $\boldsymbol{g}$ and the data. □

Corollary 2 (Monotonicity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 4$ and $I \subset [c - L/2, c + L/2]$. Set $h_1 = -\phi_{2,L} - \phi_{4,L}/4$, $\boldsymbol{g} = (g_0, g_1, g_3, \ldots, g_m)^T$, and $f_{\boldsymbol{g}} = g_0 + \sum_{j=1}^{m} g_j \phi_{j,L}$, where $g_2 = 0$, $g_4 = a_4 - a_2/4$, and $g_j = a_j$ for $j \notin \{2, 4\}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_2)h_1$, with inverse obtained by reinserting a_2 and setting $a_4 = g_4 + a_2/4$.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset [-1/\sqrt{2}, 1/\sqrt{2}]$, define

$$P_{\boldsymbol{g}}(w) = \sum_{\substack{1 \leq j \leq m \\ j \neq 2}} j g_j T_j(w), \qquad D(w) = 1 + 4w^2 - 8w^4. \tag{9}$$

Since $D \geq 1$ on J, the threshold is

$$\tau_1(\boldsymbol{g}) = \max_{w \in J} \frac{-P_{\boldsymbol{g}}(w)}{D(w)}. \tag{10}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_2$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}P_{\boldsymbol{g}}}{\mathrm{d}w} D - P_{\boldsymbol{g}} \frac{\mathrm{d}D}{\mathrm{d}w}$. This stationary polynomial has degree at most $m + 3$ when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the quotient $-P_{\boldsymbol{g}}/D$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

Proof of Corollary 2. Use the finite Gaussian model in Section 2 with the rank and closed padding interval in Corollary 2. Substitution of the inverse map proves the curve identity and invertibility, while the identity $T_j(w(x)) = \cos(j\theta(x))$ gives the derivative formulas

$$\frac{\mathrm{d}f_{\boldsymbol{g}}}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} P_{\boldsymbol{g}}(w(x)), \qquad \frac{\mathrm{d}h_1}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} D(w(x)).$$

The assumed padding confines $\theta(x)$ to $[\pi/4, 3\pi/4]$, where the cosine is strictly decreasing and $D(w) = 1 + 4w^2(1 - 2w^2) \geq 1$, including at the physical endpoints. The positive derivative bound therefore allows Theorem 1 to supply the conditioned law and strictness. Because the map transports the original Gaussian covariance, g_4 and a_2 retain their induced dependence.

Since $D > 0$ on compact J, the extreme value theorem and Fermat's theorem locate the maximum of $-P_{\boldsymbol{g}}/D$ at an endpoint or an interior stationary point. The quotient rule gives the stationary polynomial in the statement, whose degree is at most $m + 3$ because $P_{\boldsymbol{g}}$ and D have degrees at most m and 4. If this polynomial vanishes identically, the quotient is constant by the mean value theorem and either endpoint suffices. Otherwise it has finitely many distinct roots, each a candidate when it lies in the interior of J, regardless of multiplicity. This also covers reduced degrees, including a nonzero constant stationary polynomial with no interior candidates. □

Corollary 3 (Convexity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 2$ and $I \subset (c - L, c + L)$. Set $h_2 = -\phi_{1,L}$, $\boldsymbol{g} = (a_0, a_2, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = a_0 + \sum_{j=2}^{m} a_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_1)h_2$, with inverse obtained by reinserting a_1.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define the polynomial and threshold

$$Q_{\boldsymbol{g}}(w) = \sum_{j=2}^{m} j^2 a_j U_{j-1}(w), \qquad \tau_2(\boldsymbol{g}) = \max_{w \in J} Q_{\boldsymbol{g}}(w). \tag{11}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_1$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}Q_{\boldsymbol{g}}}{\mathrm{d}w}$. The polynomial $Q_{\boldsymbol{g}}$ has degree at most $m-1$, so its stationary polynomial has degree at most $m-2$ when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, $Q_{\boldsymbol{g}}$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

Proof of Corollary 3. Use the finite Gaussian model in Section 2 with the rank and open padding interval in Corollary 3. The coordinate map is an invertible signed permutation. Since $0 < \theta(x) < \pi$ on compact I,

$$\frac{\mathrm{d}^2 h_2}{\mathrm{d}x^2}(x) = \frac{\omega_1^2}{\sqrt{L}} \sin\theta(x) > 0, \qquad \frac{\mathrm{d}^2 f_{\boldsymbol{g}}}{\mathrm{d}x^2}(x) = -\frac{\omega_1^2}{\sqrt{L}} \sin\theta(x)\, Q_{\boldsymbol{g}}(w(x)).$$

The second formula uses $\sin(j\theta(x)) = \sin\theta(x)\, U_{j-1}(w(x))$, while compactness keeps the positive curvature in the first formula bounded away from zero. Thus Theorem 1 gives the claimed threshold and posterior under the stated open padding condition.

The extreme value theorem and Fermat's theorem locate the maximum of $Q_{\boldsymbol{g}}$ at an endpoint or an interior zero of $\frac{\mathrm{d}Q_{\boldsymbol{g}}}{\mathrm{d}w}$. The degree of $Q_{\boldsymbol{g}}$ is at most $m-1$, so this derivative has degree at most $m-2$ when nonzero. If it vanishes identically, $Q_{\boldsymbol{g}}$ is constant and either endpoint suffices. Otherwise its finitely many distinct interior roots enter the candidate set regardless of multiplicity, with none when the derivative is a nonzero constant. The same argument covers every reduced degree. □

Proposition 1 (Joint sign constraints)**.** *Under the finite Gaussian model in Section 2, let $\boldsymbol{a} = (a_0, \ldots, a_m)^T$ denote the coefficient vector and fix a nonempty $A \subseteq \{0,1,2\}$ and signs $\varepsilon : A \to \{-1,1\}$, with event $\mathcal{C}_A^{\varepsilon}$ defined in Equation (12). Suppose an invertible linear change of coordinates $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ gives $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h$. Here h is a fixed function in the retained span, shared by all selected orders, γ is its scalar coefficient, and $f_{\boldsymbol{g}}$ depends linearly on the remaining coordinates $\boldsymbol{g}$. Assume $h, f_{\boldsymbol{g}} \in C^{\max A}(I)$ with $\min_{x\in I} \varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x) > 0$ for every $r \in A$. Define*

$$\eta_r(x;\boldsymbol{g}) = \frac{-\varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x)}, \qquad \tau_A^{\varepsilon}(\boldsymbol{g}) = \max_{r\in A} \max_{x\in I} \eta_r(x;\boldsymbol{g}). \tag{13}$$

This finite, attained threshold satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_A^{\varepsilon}$ exactly when $\gamma \geq \tau_A^{\varepsilon}(\boldsymbol{g})$. The event has positive ordinary posterior probability and a null coefficient boundary. Its marginal density and reconstruction are Equations (6) and (7), with $\tau_r, \mathcal{C}_r$ replaced by $\tau_A^{\varepsilon}, \mathcal{C}_A^{\varepsilon}$ and the Gaussian parameters in Equation (4) computed under this map. The conditioned law satisfies $\min_{x\in I} \varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r}(x) > 0$ almost surely for every $r \in A$. The envelope, strong log-concavity, and conservative threshold-error conclusions extend under their respective hypotheses.

Proof of Proposition 1. Fix the coefficient map and signed direction satisfying the hypotheses of Proposition 1. Dividing each signed derivative inequality by its positive direction derivative gives $\gamma \geq \eta_r(x;\boldsymbol{g})$, so their intersection gives Equation (13). Compactness of I, finiteness of A, and the denominator bounds make the maximum finite and attained. The Gaussian conditioning argument in the proof of Theorem 1 then gives positive probability, the marginal density, and reconstruction.

Above the threshold, every selected signed derivative has a strictly positive minimum. At equality, an attained maximizing pair gives a zero derivative, and perturbing γ crosses the event boundary. Thus the boundary is the threshold graph, which has zero conditional Gaussian probability for every $\boldsymbol{g}$, establishing almost-sure strictness. Each $\eta_r(x;\boldsymbol{g})$ is linear in $\boldsymbol{g}$, so the envelope and strong log-concavity arguments apply with the compact index set $A \times I$, and Proposition 2 applies to the same scalar truncation. □

A.2 Conservative threshold error

Proposition 2 (Posterior error from a conservative continuum threshold)**.** *Under Theorem 1, let $\widetilde{\tau}_r$ be measurable and satisfy $\tau_r \le \widetilde{\tau}_r \le \tau_r + \delta$ almost everywhere under the ordinary posterior of $\boldsymbol{g}$, for a fixed $\delta \ge 0$. Use the same upper threshold in marginal weighting and reconstruction. Let $P_{\boldsymbol{y}}$ be the ordinary coefficient posterior, and use $\mathcal{C}_r$ also for its coefficient preimage. Define $\widetilde{\mathcal{C}}_r = \{\gamma \ge \widetilde{\tau}_r(\boldsymbol{g})\}$. Both conditioned laws exist and, with $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$,*

$$\mathrm{TV}(P_{\boldsymbol{y}}(\cdot \mid \widetilde{\mathcal{C}}_r), P_{\boldsymbol{y}}(\cdot \mid \mathcal{C}_r)) = 1 - \frac{P_{\boldsymbol{y}}(\widetilde{\mathcal{C}}_r)}{P_{\boldsymbol{y}}(\mathcal{C}_r)} \le \min\left(1, \frac{\delta}{s\sqrt{2\pi} P_{\boldsymbol{y}}(\mathcal{C}_r)}\right). \tag{15}$$

Every measurable summary of the reconstructed coefficients inherits this bound. The identity holds for any inner event of positive probability.

Proof of Proposition 2. Fix the ordinary Gaussian law, selected order, and almost-everywhere error condition of Proposition 2, modifying the numerical threshold on the exceptional null set if necessary. Finite thresholds and $s > 0$ give positive conditional tails, so both conditioning events have positive probability. To compare the resulting laws, put $A = \mathcal{C}_r$ and $B = \widetilde{\mathcal{C}}_r$. If the omitted mass is zero, both conditioned laws agree. Otherwise the law on A is a mixture of the laws on B and $A \setminus B$, with weights $P_{\boldsymbol{y}}(B)/P_{\boldsymbol{y}}(A)$ and its complement. The components have disjoint supports, so total variation equals the second weight, establishing the identity for any positive-probability inner event.

For the threshold events considered here, conditional Gaussian integration identifies the omitted mass as

$$\mathrm{E}\left[\Phi((\widetilde{\tau}_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s) - \Phi((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s) \mid \boldsymbol{y}\right].$$

Since the conditional density is at most $1/(s\sqrt{2\pi})$ and the excluded interval has length at most δ, dividing this mass bound by the probability of the exact event proves Equation (15). The invertible coefficient map preserves total variation, while measurable summaries cannot increase it, proving the remaining assertion. □

A.3 Conditioned posterior approximation

Theorem 2 (Convergence of shape-constrained HSGP posteriors)**.** *Fix $r \in \{0,1,2\}$ under Assumptions 1 and 2, and consider finite models with $L \ge 2R$ and $m \ge 1$ for $r = 0$, $m \ge 4$ for $r = 1$, or $m \ge 2$ for $r = 2$. The underlying GP has a $C^r(I)$ version, and the finite and underlying shape events have positive probability and null boundaries in that space. Consequently, the posteriors conditional on the data and $\mathcal{C}_r$ exist, with finite moments of every polynomial order in $\|\cdot\|_{C^r}$. As $L \to \infty$ and $m/L \to \infty$,*

$$W_2(\Pi^{\mathcal{C}_r}_{m,L,\boldsymbol{y}}, \Pi^{\mathcal{C}_r}_{\boldsymbol{y}}) \longrightarrow 0. \tag{16}$$

This convergence also holds for the posterior summaries: means of $\frac{\mathrm{d}^j f}{\mathrm{d}x^j}$ converge uniformly on I for every $0 \le j \le r$, and covariances between $\frac{\mathrm{d}^{r_1} f}{\mathrm{d}x^{r_1}}(x)$ and $\frac{\mathrm{d}^{r_2} f}{\mathrm{d}x^{r_2}}(x')$ converge uniformly on $I \times I$ for every $0 \le r_1, r_2 \le r$. Under the conditioned underlying law, $\min_I \frac{\mathrm{d}^r f}{\mathrm{d}x^r} > 0$ almost surely.

Proof of Theorem 2. Fix r and any sequence of models satisfying the conditions and limits in Theorem 2. Write $F_{m,L}$ for a draw from the finite prior and $k_{m,L}(x,x') = \sum_{j=1}^m S_k(\omega_j)\phi_{j,L}(x)\phi_{j,L}(x')$ for the covariance of its spectral component. We first establish weak convergence of the priors in $C^r(I)$ and uniform norm moments. Positive shape probability and a boundary of prior probability zero will then allow passage to the conditioned posteriors, whose moment bounds give Wasserstein convergence and convergence of the stated summaries.

Covariance and increment bounds. Put $b = 2r + 2\zeta$. The two spectral weights in Assumption 2 also control the intermediate orders needed for covariance derivatives. For each integer $0 < d \le 2r$, differentiating $\omega^d S_k(\omega)$ on $[0,1]$ gives integrable terms because S_k is bounded there and its derivative is integrable. On $[1,\infty)$, write the same function as $\omega^{d-b}(\omega^b S_k(\omega))$. The multiplier and its derivative are bounded, so the product and its derivative are integrable. Thus, for $d = 0,\dots,2r$ and $d = b$, the weighted spectra are locally absolutely continuous and have themselves and their derivatives in $L^1(0,\infty)$, with the cases $d = 0$ and $d = b$ supplied directly by the assumption. Their finite values at zero and integrable derivatives also give boundedness. The derivative-covariance argument of Jensen and Hoffmann (2026, Theorem 2 and its proof) then gives, for $0 \le r_1, r_2 \le r$, with $d = r_1 + r_2$:

$$\sup_{x,x' \in I} \left| \frac{\partial^{r_1+r_2}(k_{m,L}(x,x') - k(x-x'))}{\partial x^{r_1} \partial (x')^{r_2}} \right| \le \frac{b_d}{L} + \frac{1}{\pi} \int_{\pi m/(2L)}^{\infty} \omega^d S_k(\omega)\, d\omega, \tag{18}$$

where $b_d < \infty$ is independent of m and L. The cited theorem states stronger conditions on classical derivatives, but its proof bounds frequency-cell errors by integrals of absolute derivatives and the reflected term by consecutive frequency differences. Both steps require only the absolute continuity, integrability, and boundedness verified above. In Equation (18), the first term vanishes as the padded boundary recedes, while the second vanishes as the frequency cutoff increases. The common intercept adds no covariance error.

For the frequency spacing $\Delta = \pi/(2L)$, with $\omega_j = j\Delta$, the same frequency-cell estimate bounds the spectral sums by

$$\Delta \sum_{j=1}^{m} (j\Delta)^d S_k(j\Delta) \le \|\omega^d S_k(\omega)\|_1 + \Delta \left\| \frac{\mathrm{d}(\omega^d S_k(\omega))}{\mathrm{d}\omega} \right\|_1, \qquad d \in \{2j : 0 \le j \le r\} \cup \{b\}.$$

Here the norms are over $[0,\infty)$. The right-hand side is uniformly bounded because $L \ge 2R$ gives $\Delta \le \pi/(4R)$. Taking $d = 0, 2, \dots, 2r$ therefore bounds the derivative variances at x_-, with the intercept adding its fixed variance at order zero.

To control increments of the highest derivative, we use this estimate with $d = b = 2r + 2\zeta$. Each differentiated basis function is a sine or cosine with amplitude $L^{-1/2}\omega_j^r$, so $|\sin u - \sin v| \le 2^{1-\zeta}|u-v|^\zeta$ and the corresponding cosine inequality give

$$\left| \frac{\mathrm{d}^r \phi_{j,L}}{\mathrm{d}x^r}(x) - \frac{\mathrm{d}^r \phi_{j,L}}{\mathrm{d}x^r}(x') \right| \le \frac{2^{1-\zeta}}{\sqrt{L}} \omega_j^{r+\zeta} |x - x'|^\zeta.$$

By independence of the Gaussian coefficients, the variance of the rth derivative increment is the sum of the squared basis increments above weighted by $S_k(\omega_j)$. The intercept cancels in increments when $r = 0$ and contributes no derivative when $r \ge 1$. Consequently,

$$\mathrm{Var}\left(\frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}(x) - \frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}(x') \right) \le C_\zeta |x - x'|^{2\zeta} \Delta \sum_{j=1}^{m} \omega_j^{2r+2\zeta} S_k(\omega_j),$$

where $C_\zeta = 2^{3-2\zeta}/\pi$ follows from $1/L = 2\Delta/\pi$. The frequency-cell estimate at $d = b$ bounds the spectral sum on the right uniformly in m, L, giving the increment bound $A|x - x'|^{2\zeta}$ with A independent of rank and padding. Fourier inversion gives the same bound for the underlying mean-square rth derivative, after increasing A if needed. Gaussian moments then verify the Kolmogorov continuity criterion (Kallenberg, 2021) with any exponent p satisfying $p\zeta > 1$. For $r = 0$, this gives a continuous version of W. For $r \ge 1$, integrating the continuous highest derivative, with the lower mean-square derivative values at x_- as integration constants, recovers a version of W by the mean-square fundamental theorem of calculus. Adding the intercept therefore gives a $C^r(I)$ version of F.

Tightness and norm moments. We next control both the size and the continuity of the sample paths uniformly over the finite models. These bounds establish tightness in $C^r(I)$ and provide the norm moments needed later for Wasserstein convergence and convergence of posterior summaries. Let H denote either $\frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}$ or $\frac{\mathrm{d}^r F}{\mathrm{d}x^r}$, and choose $p > \max(4, 1/\zeta)$ and a Hölder exponent $0 < \kappa < \zeta - 1/p$. For $j \geq 0$, let M_j be the largest absolute increment of H between neighboring points on the mesh dividing I into 2^j equal cells. The Gaussian increment bound gives $\mathrm{E}[M_j^p]^{1/p} \leq C_p 2^{-(\zeta-1/p)j}$ uniformly over the models, because the pth power of the maximum is bounded by the sum of the pth powers of the 2^j increments. Approximating arbitrary $x, x' \in I$ by points on successively finer meshes expresses $H(x) - H(x')$ through adjacent increments. This dyadic chaining argument bounds the Hölder seminorm by

$$[H]_\kappa = \sup_{\substack{x,x' \in I \\ x \neq x'}} \frac{|H(x) - H(x')|}{|x - x'|^\kappa} \leq C_{I,\kappa} \sum_{j=0}^{\infty} 2^{\kappa j} M_j.$$

Here $C_{I,\kappa}$ depends only on I and κ. Minkowski's inequality therefore bounds $\mathrm{E}[[H]_\kappa^p]^{1/p}$ by a constant times $\sum_{j \geq 0} 2^{-(\zeta-1/p-\kappa)j}$. Our choice of κ makes this series converge, giving a uniform pth moment bound for the seminorm.

The seminorm and the value $H(x_-)$ together bound the highest derivative everywhere on I. When $r \geq 1$, repeated integration, starting from the lower derivative values at x_-, then bounds every lower derivative and the curve itself. For $r = 0$, the first bound already controls the curve. The uniformly bounded Gaussian moments of these endpoint values consequently give

$$\sup_{m,L} \mathrm{E}[\|F_{m,L}\|_{C^r}^p] + \mathrm{E}[\|F\|_{C^r}^p] < \infty \tag{19}$$

over the permitted ranks and padding. Since p can be arbitrarily large, every polynomial norm moment is uniformly bounded.

Bounds on derivative values of orders $0, \ldots, r$ at x_- and on the highest derivative's Hölder seminorm define relatively compact subsets of $C^r(I)$ by Arzelà–Ascoli. The moment bounds and Markov's inequality allow these bounds to be chosen so that the finite priors assign arbitrarily high probability to such a set, uniformly over the models. This proves tightness. To identify the limit, the common mean and Equation (18) give convergence of the joint Gaussian laws of every finite collection of derivative values. Such evaluations on a countable dense subset of I determine a $C^r(I)$ law, so every subsequential limit is the underlying prior. The finite priors therefore converge weakly to that law.

Positive shape probability and a null boundary. The boundary of $\mathcal{C}_r$ in $C^r(I)$ consists exactly of curves with $\min_I \frac{\mathrm{d}^r f}{\mathrm{d}x^r} = 0$. Indeed, this minimum is continuous in the C^r norm, and adding or subtracting a small multiple of $(x - c)^r/r!$ moves any such curve to either side of the event. We therefore need positive probability of nonnegativity and zero probability of equality. For each order, these follow by separating one Gaussian coordinate whose contribution to the selected derivative is positive throughout I.

For $r = 0$, the independent intercept gives

$$\min_{x \in I} F(x) = a_0 + \min_{x \in I} W(x).$$

Conditional on W, nonnegativity therefore imposes a finite lower threshold on the nondegenerate Gaussian a_0, giving positive conditional probability and zero conditional probability of equality.

For $r \in \{1, 2\}$, the intercept contributes no derivative. We instead seek a decomposition of W into a Gaussian coefficient multiplying a deterministic function whose rth derivative is uniformly positive, together with an independent Gaussian process. This reduces the shape restriction to a scalar threshold, as in the finite model. The reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ of W in $C^r(I)$ identifies the deterministic directions that can be separated in this way.

To construct a candidate direction, choose $0 < \omega_- < \omega_+ < \pi/(2R)$ and set

$$v_r(x) = \int_{\omega_-}^{\omega_+} S_k(\omega) \cos(\omega(x - c) - r\pi/2)\, d\omega.$$

Taking r derivatives removes the phase shift, leaving the cosine factor $\cos(\omega(x - c))$. Because $|x - c| \le R$ and $\omega_+ R < \pi/2$, this factor is uniformly positive on the selected frequency band and throughout I. Thus

$$\frac{\mathrm{d}^r v_r}{\mathrm{d}x^r}(x) \ge \cos(\omega_+ R) \int_{\omega_-}^{\omega_+} \omega^r S_k(\omega)\, d\omega > 0.$$

We next verify that this direction belongs to $\mathcal{H}$, which justifies separating a Gaussian coefficient along it. The Fourier expression for the full-line RKHS norm gives squared norm $\pi \int_{\omega_-}^{\omega_+} S_k(\omega)\, d\omega < \infty$ for v_r, and restricting the function to I cannot increase that norm. The RKHS obtained from the covariance on I agrees with the RKHS of W as a $C^r(I)$-valued Gaussian random element by van der Vaart and van Zanten (2008, Lemma 8.1), applied to the continuous, injective embedding into $C(I)$. The norm calculation therefore establishes $v_r \in \mathcal{H}$ in the function space used in this proof.

Since v_r is nonzero, we can normalize it by its RKHS norm. The Gaussian RKHS isometry (van der Vaart and van Zanten, 2008, Lemma 2.3) then supplies a standard normal variable ξ, jointly Gaussian with W, such that $\mathrm{E}[\xi W(x)] = v_r(x)/\|v_r\|_{\mathcal{H}}$ for every $x \in I$. Define the remaining process V by

$$W = \xi \frac{v_r}{\|v_r\|_{\mathcal{H}}} + V.$$

Subtracting this component preserves $C^r(I)$ paths, and $\mathrm{Var}(\xi) = 1$ gives $\mathrm{Cov}(\xi, V(x)) = 0$ for every $x \in I$. Joint Gaussianity therefore makes ξ independent of the whole process V.

Conditional on V, only the scalar coefficient ξ remains random in the derivative constraint. Dividing each derivative inequality by the uniformly positive derivative of v_r gives

$$\{F \in \mathcal{C}_r\} = \left\{ \xi \ge \|v_r\|_{\mathcal{H}} \max_{x \in I} \frac{-\frac{\mathrm{d}^r V}{\mathrm{d}x^r}(x)}{\frac{\mathrm{d}^r v_r}{\mathrm{d}x^r}(x)} \right\}.$$

For each path of V the threshold is finite, so the conditional normal tail has positive probability and no atom at equality. Equality is precisely the zero-minimum event identified above, so the underlying shape event has positive probability and a null boundary at all three orders. The finite laws have the same properties by Corollaries 1 to 3, applied with likelihood one. In both cases, conditioning on nonnegativity therefore gives a strictly positive minimum selected derivative almost surely.

Likelihood and shape conditioning. Write $\mathcal{L}_{\boldsymbol{y}}(f)$ for the Gaussian density at $\boldsymbol{y}$ with mean $(f(x_1), \ldots, f(x_n))^T$ and covariance $\boldsymbol{\Sigma}$, taking $\mathcal{L}_{\boldsymbol{y}}(f) = 1$ when $n = 0$. This likelihood is bounded, positive, and continuous on $C^r(I)$. For any bounded continuous test function $\psi : C^r(I) \to \mathbb{R}$, the map

$$f \longmapsto \psi(f) \mathcal{L}_{\boldsymbol{y}}(f) \mathbf{1}_{\mathcal{C}_r}(f)$$

is bounded and continuous outside $\partial \mathcal{C}_r$, where $\mathbf{1}_{\mathcal{C}_r}(f)$ is one if $f \in \mathcal{C}_r$ and zero otherwise. The preceding argument shows that this boundary has zero probability under the underlying prior, so

the Portmanteau theorem (Billingsley, 1999, Section 2) gives convergence of the integrals of this map under the finite priors to its integral under the underlying prior. Taking $\psi \equiv 1$ shows that the normalizing constants converge to a positive limit. Dividing by them gives convergence of posterior expectations of every bounded continuous ψ, which is conditioned weak convergence. The posterior densities relative to the priors are uniformly bounded along the sequence by the likelihood supremum divided by these constants, so Equation (19) also holds for the conditioned laws at every fixed polynomial order. Likelihood weighting preserves the zero boundary probability, giving the strictness assertion.

Wasserstein convergence and posterior summaries. We now strengthen conditioned weak convergence to convergence in Wasserstein distance. Index the chosen approximation sequence by ℓ, with ranks m_ℓ and padding L_ℓ. In this paragraph, write F_ℓ and F for draws from the finite and underlying conditioned posteriors, respectively. Since $C^r(I)$ is separable and complete, the Skorokhod representation theorem (Billingsley, 1999, Theorem 6.7) allows these draws to be constructed on a common probability space, preserving their posterior laws, so that $\|F_\ell - F\|_{C^r} \to 0$ almost surely. The uniform fourth-moment bounds established above make the squared norm differences uniformly integrable, so their expectations also converge to zero. Wasserstein distance takes the infimum over all couplings, and this particular coupling therefore gives

$$W_2^2(\Pi^{\mathcal{C}^r}_{m_\ell, L_\ell, \boldsymbol{y}}, \Pi^{\mathcal{C}^r}_{\boldsymbol{y}}) \le \mathrm{E}[\|F_\ell - F\|^2_{C^r}] \longrightarrow 0,$$

proving Equation (16).

The same coupling controls posterior summaries uniformly in their input locations because the C^r norm bounds every derivative through order r over the whole interval. For each such order, the uniform difference of derivative means is at most $\mathrm{E}[\|F_\ell - F\|_{C^r}]$, which tends to zero by Cauchy–Schwarz. For any two derivative orders through r, expand the difference of their products by changing one factor at a time. The resulting difference of mixed second moments is bounded, uniformly on $I \times I$, by

$$\mathrm{E}[\|F_\ell - F\|_{C^r}(\|F_\ell\|_{C^r} + \|F\|_{C^r})].$$

Cauchy–Schwarz bounds this expectation by the product of $\mathrm{E}[\|F_\ell - F\|^2_{C^r}]^{1/2}$, which tends to zero, and $\mathrm{E}[(\|F_\ell\|_{C^r} + \|F\|_{C^r})^2]^{1/2}$, which remains bounded. Thus the mixed second moments converge uniformly. Subtracting the uniformly convergent products of means proves covariance convergence. Since the sequence was arbitrary, all these conclusions hold in the full stated limiting regime. □

A.4 Signed combinations in the Laplacian expansion

Corollary 4 (Signed combinations and their reductions)**.** *Under the finite Gaussian model in Section 2, let $A \subseteq \{0,1,2\}$ be nonempty and fix signs $\varepsilon_r \in \{-1,1\}$ for $r \in A$. For $\theta(x) = \omega_1(x-c+L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1,1)$, define, for a coefficient array $\boldsymbol{b} = (b_0, \dots, b_m)^T$,*

$$H_{0,\boldsymbol{b}}(w) = \sqrt{L}\, b_0 + \sqrt{1-w^2} \sum_{j=1}^{m} b_j U_{j-1}(w),$$

$$H_{1,\boldsymbol{b}}(w) = \sum_{j=1}^{m} j b_j T_j(w), \qquad H_{2,\boldsymbol{b}}(w) = -\sum_{j=1}^{m} j^2 b_j U_{j-1}(w).$$

Suppose a fixed $\boldsymbol{d} = (d_0, \dots, d_m)^T$ satisfies $\min_{w \in J} \varepsilon_r H_{r,\boldsymbol{d}}(w) > 0$ for every $r \in A$. Set $h = d_0 + \sum_{j=1}^m d_j \phi_{j,L}$ and choose $j_* \in \{0, \dots, m\}$ with $d_{j_*} \neq 0$. The coordinates

$$\gamma = \frac{a_{j_*}}{d_{j_*}}, \qquad g_j = a_j - d_j\gamma \quad (j \neq j_*), \qquad g_{j_*} = 0$$

give $\boldsymbol{g} = (g_j)_{j \neq j_*} \in \mathbb{R}^m$ *and* $f_{\boldsymbol{g}} = g_0 + \sum_{j=1}^m g_j \phi_{j,L}$. *Then* $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h$, *with inverse* $a_j = g_j + d_j \gamma$ *for* $0 \le j \le m$. *In expressions involving* $H_{r,\boldsymbol{g}}$, *the zero coordinate* g_{j_*} *is included in the coefficient array.*

With η_r *from Equation* (13), *define*

$$\eta_r(x; \boldsymbol{g}) = -\frac{H_{r,\boldsymbol{g}}(w(x))}{H_{r,\boldsymbol{d}}(w(x))}, \qquad M_r(\boldsymbol{g}) = \max_{x \in I} \eta_r(x; \boldsymbol{g}), \qquad \eta_r^{\pm}(\boldsymbol{g}) = \eta_r(x_{\pm}; \boldsymbol{g}).$$

For adjacent selected orders $r, r+1 \in A$, *let* $E_r(\boldsymbol{g}) = \eta_r^-(\boldsymbol{g})$ *when* $\varepsilon_r = \varepsilon_{r+1}$ *and* $E_r(\boldsymbol{g}) = \eta_r^+(\boldsymbol{g})$ *otherwise. Suppressing the argument* $\boldsymbol{g}$ *on the right, the threshold is*

$$\tau_A^{\varepsilon}(\boldsymbol{g}) = \begin{cases} M_r, & A = \{r\}, \\ \max\{M_1, E_0\}, & A = \{0,1\}, \\ \max\{M_2, M_0\}, & A = \{0,2\},\ \varepsilon_0 = \varepsilon_2, \\ \max\{M_2, \eta_0^-, \eta_0^+\}, & A = \{0,2\},\ \varepsilon_0 \neq \varepsilon_2, \\ \max\{M_2, E_1\}, & A = \{1,2\}, \\ \max\{M_2, E_1, E_0\}, & A = \{0,1,2\}. \end{cases}$$

Proposition 1 applies with scalar coefficient $\gamma = a_{j_*}/d_{j_*}$. *Thus* $f_{\boldsymbol{a}} \in \mathcal{C}_A^{\varepsilon}$ *exactly when* $\gamma \ge \tau_A^{\varepsilon}(\boldsymbol{g})$.

Each maximum M_r *is attained at a physical endpoint or a location whose cosine coordinate is an interior root in* J *of*

$$H_{r,\boldsymbol{d}}(w) \frac{\mathrm{d}H_{r,\boldsymbol{g}}}{\mathrm{d}w}(w) - H_{r,\boldsymbol{g}}(w) \frac{\mathrm{d}H_{r,\boldsymbol{d}}}{\mathrm{d}w}(w) = 0.$$

For $r = 1, 2$, *the left side is a polynomial. All root multiplicities and reduced degrees are included. If the left side vanishes identically, the quotient is constant and either endpoint suffices.*

These formulas cover the 26 nonempty combinations without opposing signs, with positive ordinary posterior probability whenever the stated direction condition holds for the selected combination. The remaining 37 selections contain opposing signs. At orders zero, one, and two these force, respectively, a zero, constant, or affine function, subject to any further restrictions. In the finite expansion, affine curves are constant, so all 37 events have ordinary posterior probability zero.

Proof. Fix the selected orders, signs, and direction coefficients satisfying Corollary 4. The inverse map follows by substitution, so the coordinates preserve the full coefficient space. The Chebyshev identities in Definition 1 give

$$\begin{aligned} f_{\boldsymbol{b}}(x) &= \frac{H_{0,\boldsymbol{b}}(w(x))}{\sqrt{L}}, \\ \frac{\mathrm{d}f_{\boldsymbol{b}}}{\mathrm{d}x}(x) &= \frac{\omega_1}{\sqrt{L}} H_{1,\boldsymbol{b}}(w(x)), \\ \frac{\mathrm{d}^2 f_{\boldsymbol{b}}}{\mathrm{d}x^2}(x) &= \frac{\omega_1^2 \sin\theta(x)}{\sqrt{L}} H_{2,\boldsymbol{b}}(w(x)). \end{aligned}$$

Since $\sin\theta$ is positive on compact I, the assumed conditions on $H_{r,\boldsymbol{d}}$ give the uniform signed derivative bounds required by Proposition 1. The positive factors cancel from the ratios, yielding the displayed expression for η_r and the conditioned posterior.

A signed slope restriction makes the minimum of $\varepsilon_0 f_{\boldsymbol{a}}$ occur at x_- when $\varepsilon_0 = \varepsilon_1$, and at x_+ otherwise. This gives E_0. Applying the same argument to the signed slope $\varepsilon_1 \frac{\mathrm{d}f_{\boldsymbol{a}}}{\mathrm{d}x}$ under the curvature restriction gives E_1. When $\varepsilon_0 \neq \varepsilon_2$, the signed curve $\varepsilon_0 f_{\boldsymbol{a}}$ is concave, so its nonnegativity is

equivalent to nonnegative endpoint values by the chord inequality. When the signs agree, the signed curve is convex and its minimum may be interior, so the value restriction retains M_0. Together with the continuum slope and curvature bounds M_1 and M_2, these equivalences give every displayed threshold, because each pointwise restriction is $\gamma \geq \eta_r(x; \boldsymbol{g})$.

The cosine coordinate is strictly decreasing on I, and the signed denominators stay positive on compact J. The extreme value theorem, Fermat's theorem, and the quotient rule therefore give the stated endpoint and stationary candidates. The functions $H_{r,\boldsymbol{b}}$ are analytic on $(-1, 1)$, so a nonzero stationary expression has only finitely many roots on J. If it vanishes identically, the mean value theorem gives constancy of the quotient.

Choosing no restriction or one of two signs at each order gives $3^3 - 1 = 26$ nonempty combinations: six singles, twelve pairs, and eight triples. Of the $2^6 - 1 = 63$ nonempty selections from the six constraints, the other 37 contain opposing signs, which force the corresponding derivative to vanish identically. This gives zero, a constant, or an affine function. Analyticity extends any affine identity on I to the padded interval, whose two endpoint values both equal a_0, so the affine curve is constant. Linear independence of the sine basis gives $a_1 = \cdots = a_m = 0$, with $a_0 = 0$ as well when both value signs are imposed. These proper linear subspaces have zero probability under the nondegenerate Gaussian posterior. □

Table 4: Implementing the 20 joint combinations without opposing constraints under Corollary 4. P and N denote nonnegativity and nonpositivity, I and D denote nondecreasing and nonincreasing curves, and CX and CV denote convexity and concavity. Choose fixed direction coefficients $\boldsymbol{d}$ so that every signed profile in the second column is strictly positive throughout J. For a pivot $d_{j_*} \neq 0$, use $\gamma = a_{j_*}/d_{j_*}$ and $g_j = a_j - d_j\gamma$, so that $g_{j_*} = 0$ and $a_j = g_j + d_j\gamma$ for $0 \leq j \leq m$. Every displayed coefficient bound in a row must hold, giving $\gamma \geq \tau_A^{\varepsilon}(\boldsymbol{g})$ by taking the largest displayed right-hand side. With arguments $\boldsymbol{g}$ suppressed, $M_r = \max_{x \in I} \eta_r(x; \boldsymbol{g})$ requires an interval maximum, evaluated at the endpoint and stationary candidates in Corollary 4, whereas $\eta_r^{\pm} = \eta_r(x_{\pm}; \boldsymbol{g})$ requires only evaluation at the left endpoint x_- or right endpoint x_+. All ratios $\eta_r(x; \boldsymbol{g}) = -H_{r,\boldsymbol{g}}(w(x))/H_{r,\boldsymbol{d}}(w(x))$ use the same direction and coefficient map within a row. A dash indicates no separate endpoint bound.

Combination	Direction condition on J	Continuum bound	Endpoint bound
Pairs of constraints			
P, I	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}} > 0$	$\gamma \geq M_1$	$\gamma \geq \eta_0^-$
P, D	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}} > 0$	$\gamma \geq M_1$	$\gamma \geq \eta_0^+$
N, I	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}} > 0$	$\gamma \geq M_1$	$\gamma \geq \eta_0^+$
N, D	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}} > 0$	$\gamma \geq M_1$	$\gamma \geq \eta_0^-$
P, CX	$H_{0,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq \max\{M_2, M_0\}$	—
P, CV	$H_{0,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_0^-, \eta_0^+\}$
N, CX	$-H_{0,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_0^-, \eta_0^+\}$
N, CV	$-H_{0,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq \max\{M_2, M_0\}$	—
I, CX	$H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \eta_1^-$
I, CV	$H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \eta_1^+$
D, CX	$-H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \eta_1^+$
D, CV	$-H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \eta_1^-$
Triples of constraints			
P, I, CX	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^-, \eta_0^-\}$
P, I, CV	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^+, \eta_0^-\}$

Continued on the next page

Table 4 (continued)

Combination	Direction condition on J	Continuum bound	Endpoint bound
P, D, CX	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^+, \eta_0^+\}$
P, D, CV	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^-, \eta_0^+\}$
N, I, CX	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^-, \eta_0^+\}$
N, I, CV	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^+, \eta_0^+\}$
N, D, CX	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^+, \eta_0^-\}$
N, D, CV	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}} > 0$	$\gamma \geq M_2$	$\gamma \geq \max\{\eta_1^-, \eta_0^-\}$

A.5 Whitened posterior geometry

Proposition 3 (Strong log-concavity of the marginal posterior)**.** *Under Theorem 1, let* $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ *with* $\boldsymbol{B}$ *invertible and put* $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$*. The induced density* q_r *of* $\boldsymbol{z}$ *is* 1*-strongly log-concave. That is,* $\log q_r(\boldsymbol{z}) + \|\boldsymbol{z}\|^2/2$ *is concave. It has a unique mode and finite moments of every polynomial order.*

Proof. Fix the Gaussian posterior, selected order, and invertible whitening factor in Proposition 3. Since τ_r is a maximum of linear functions, it is convex. Subtracting the affine conditional mean and dividing by $s > 0$ therefore makes $\alpha_r = (\tau_r - \mu)/s$ convex in the whitened coordinates. Under the ordinary posterior, $\eta = (\gamma - \mu(\boldsymbol{g}))/s$ is standard normal and independent of $\boldsymbol{z}$. The joint standard normal density in $(\boldsymbol{z}, \eta)$ restricted to the convex epigraph $\eta \geq \alpha_r(\boldsymbol{z})$ is 1-strongly log-concave and has positive normalizer by Theorem 1. Preservation under marginalization (Saumard and Wellner, 2014, Theorem 3.8) gives the same constant for q_r. To obtain the moment and mode conclusions, use the survival factor's continuity, positivity, and upper bound of one: q_r is continuous and bounded above by the standard normal density divided by the positive event probability. This bound gives all polynomial moments and convergence of the density to zero at infinity, so continuity implies attainment of a mode, which strong log-concavity makes unique. □

B Numerical evaluation

The posterior construction requires three numerical operations. We first prepare the ordinary Gaussian posterior, then evaluate the shape threshold whenever the sampler proposes a coefficient vector, and finally draw the omitted coefficient conditionally on each retained vector. The Gaussian preparation is performed once, whereas threshold evaluation is repeated throughout sampling. The main numerical issue is therefore to bound a maximum over the entire constraint interval reliably and to use the same bound in the marginal density and conditional reconstruction.

B.1 Gaussian preparation and the common algorithm

The preparation separates the coefficients that will be sampled from the scalar that will be integrated out. Let $\boldsymbol{A}$ be the invertible linear map from the original coefficients to $(\boldsymbol{g}, \gamma)$ under the applicable conditions in Corollaries 1 to 3, or the common map in Corollary 4 for a joint restriction. The ordinary posterior then has mean $\boldsymbol{A}\boldsymbol{m}$ and covariance $\boldsymbol{A}\boldsymbol{V}\boldsymbol{A}^T$. Denote the mean blocks by $\boldsymbol{m}_g$ and m_γ, the marginal covariance of $\boldsymbol{g}$ by $\boldsymbol{V}_g$, the variance of γ by v_γ, and their cross-covariance by $\boldsymbol{v} = \mathrm{Cov}(\boldsymbol{g}, \gamma \mid \boldsymbol{y})$. Gaussian conditioning gives

$$\mu(\boldsymbol{g}) = m_\gamma + \boldsymbol{v}^T \boldsymbol{V}_g^{-1}(\boldsymbol{g} - \boldsymbol{m}_g), \qquad s^2 = v_\gamma - \boldsymbol{v}^T \boldsymbol{V}_g^{-1} \boldsymbol{v}. \tag{20}$$

These formulas give the conditional mean and variance of γ for any value of $\boldsymbol{g}$. To put the sampled coefficients on a common Gaussian scale, choose an invertible factor $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ and set $\boldsymbol{g}(\boldsymbol{z}) = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$. Then Equation (20) gives $\mu(\boldsymbol{g}(\boldsymbol{z})) = m_\gamma + \boldsymbol{\lambda}^T\boldsymbol{z}$, where $\boldsymbol{\lambda} = \boldsymbol{B}^{-1}\boldsymbol{v}$ is computed by a linear solve. Under the ordinary posterior, $\boldsymbol{z}$ has a standard normal distribution. The shape restriction then enters through the survival weight in Equation (14).

When computing s^2, subtracting two nearly equal terms in Equation (20) can lose precision. The equivalent formula $s^2 = (\boldsymbol{d}^T\boldsymbol{V}^{-1}\boldsymbol{d})^{-1}$, where $\boldsymbol{d}$ is the column of $\boldsymbol{A}^{-1}$ corresponding to γ, avoids this subtraction. The quadratic form is the γ diagonal entry of the transformed precision matrix and can be evaluated using the factored posterior precision without forming $\boldsymbol{V}^{-1}$ explicitly.

The ordinary Gaussian posterior can itself be prepared using matrix factorizations and linear solves. For a general noise covariance $\boldsymbol{\Sigma}$, solves with its Cholesky factor transform the observation errors to have identity covariance. We apply the same transformation to the design and to the response centered at $\boldsymbol{X}\boldsymbol{m}_0$, then scale the design columns by the prior standard deviations. In these coordinates, the prior precision is the identity. Stacking the transformed design above an identity matrix and applying a QR factorization therefore gives a triangular factor of the posterior precision. Linear solves supply the posterior mean and covariance factors, after which the full transformation $\boldsymbol{A}\boldsymbol{V}\boldsymbol{A}^T$ preserves the dependence introduced by the coefficient map.

Algorithm 1 separates the repeated marginal-density evaluation from reconstruction after sampling. Both stages use the same threshold rule, tolerance, and precision policy, so they describe the same conservative conditioning event. A gradient-based sampler also needs a vector to guide its proposals, called the force below. We obtain it from the envelope derivative in Section B.3, which differentiates the maximized objective without differentiating the numerical search for its maximizing location.

Algorithm 1 Sign-constrained marginal sampling and reconstruction. For a joint event satisfying Proposition 1, use its common coefficient map and replace τ_r by τ_A^ε throughout, including the upper threshold used in both stages.

Require: Gaussian model, event $\mathcal{C}_r$ or $\mathcal{C}_A^\varepsilon$, admissible map, interval I
Require: Absolute tolerance $\delta > 0$ and precision budget
Form the ordinary posterior and apply the invertible shape-specific map
Prepare $\boldsymbol{m}_g, \boldsymbol{B}, m_\gamma, \boldsymbol{\lambda}, s$
function MARGINAL EVALUATION($\boldsymbol{z}$)
    Set $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$
    Set $\mu = m_\gamma + \boldsymbol{\lambda}^T\boldsymbol{z}$
    Bound $\tau_r(\boldsymbol{g})$ between ℓ and u with $u - \ell \le \delta$, or report failure
    Set $\widetilde{\tau}_r = u$ and $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$
    Return $-\|\boldsymbol{z}\|^2/2 + \log \Phi^{\mathrm{c}}(\widetilde{\alpha})$ and the envelope force
end function
Use the marginal sampler to retain draws of $\boldsymbol{z}$
for each retained $\boldsymbol{z}$ **do**
    Set $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ and $\mu = m_\gamma + \boldsymbol{\lambda}^T\boldsymbol{z}$
    Evaluate the same upper-threshold rule and set $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$
    Draw $U \sim \mathrm{Uniform}(0, 1)$ independently
    Compute $\eta = -\Phi^{-1}(U\Phi^{\mathrm{c}}(\widetilde{\alpha}))$ using log probabilities
    Set $\gamma = \widetilde{\tau}_r + s(\eta - \widetilde{\alpha})$ and invert the coefficient map
    Evaluate the reconstructed curve and requested functionals
end for

B.2 Conservative threshold evaluation

The threshold is a maximum over the entire constraint interval. Evaluating its objective at finitely many locations gives a lower bound, while an upper bound must also control values between those locations. The calculation therefore maintains both bounds and refines them until their difference is within the requested tolerance.

Bounds used for conditioning. The routine takes the remaining coefficients in their declared order, the shape case and basis geometry, a positive absolute tolerance δ in coefficient units, and an arithmetic precision limit. A successful evaluation returns

$$\ell \le \tau_r(\boldsymbol{g}) \le u, \qquad u - \ell \le \delta,$$

together with an approximate maximizing location for the envelope force. Choosing $\widetilde{\tau}_r = u$ then ensures $0 \le \widetilde{\tau}_r - \tau_r \le \delta$. The resulting lower bound on γ can be slightly too large, but cannot admit a curve that violates the constraint. If the requested enclosure cannot be obtained, the routine reports failure.

Locating and evaluating extrema. For each individual constraint, the maximum occurs at an endpoint or an interior stationary location. Corollaries 1 to 3 express the interior candidates as roots of a polynomial in the cosine coordinate $w \in J$. Real-root isolation finds disjoint intervals containing every distinct real root and establishes that none have been omitted (Johansson, 2017). The represented coefficients are treated as exact rationals, allowing repeated factors to be removed before isolation while retaining roots of even multiplicity. This arithmetic also detects reduced degree without discarding small nonzero coefficients. A constant stationary polynomial requires only endpoint evaluations, including when it is identically zero.

Interval arithmetic then encloses the objective at the endpoints and over each root interval. Rounding is directed outward so that these enclosures include arithmetic error. The largest candidate upper bound gives u, while valid lower bounds come from endpoints and roots known to belong to J. Refining the candidate enclosures reduces the gap between these bounds until it is at most δ.

Roots near the boundary of J need additional care. A root interval known to lie outside J is discarded. One crossing the boundary remains an upper-bound candidate, but supplies a lower bound only after its root is shown to belong to J. The calculation can still stop without resolving that location if the overall gap is already within δ. For quotient objectives, division also requires a strictly positive interval lower bound for the denominator.

Evaluation for the individual constraints. For positivity, isolate the roots of $P_{0,\boldsymbol{g}}$ and evaluate the objective $-f_{\boldsymbol{g}}$ over the corresponding physical locations. Interval evaluation of this trigonometric expression encloses the candidate values.

For monotonicity, the objective is $-P_{\boldsymbol{g}}/D$, with stationary polynomial

$$\frac{\mathrm{d}P_{\boldsymbol{g}}}{\mathrm{d}w} D - P_{\boldsymbol{g}} \frac{\mathrm{d}D}{\mathrm{d}w}.$$

The polynomials in Equation (9) use every spectral coefficient, with $g_2 = 0$ and the intercept excluded, and satisfy $D \ge 1$ on J. The lidar implementation first bounds the equivalent trigonometric expression, retaining all harmonics, and uses root isolation as a fallback.

For convexity, the polynomial objective $Q_{\boldsymbol{g}}$ can be bounded before finding its stationary points. Map J affinely to $[0, 1]$ and express the polynomial in the Bernstein basis. These basis functions are

nonnegative and sum to one, so every polynomial value is a weighted average of the coefficients and lies between their smallest and largest values. Ball arithmetic, which represents each quantity by a midpoint and an error radius, encloses the change of basis and the cosine endpoints.

De Casteljau subdivision rewrites the polynomial on smaller intervals, with outward rounding preserving the bounds. Repeatedly subdividing the interval with the largest upper bound refines u, while endpoint and midpoint evaluations improve ℓ. If the gap still exceeds δ when the subdivision budget is reached, isolate the roots of the exact rational derivative of $Q_{\boldsymbol{g}}$ and increase arithmetic precision as needed.

Joint restrictions. For a signed joint event, use a common coefficient map satisfying Proposition 1. Every selected order then shares the same direction h and remaining coefficients $\boldsymbol{g}$, so its quotient objective must be formed from Equation (13) under this map. Subdivide I and bound each quotient on every subinterval, verifying a positive lower bound for its signed denominator before division. Evaluations at points in I supply lower bounds.

If $[\ell_r, u_r]$ encloses the maximum for order $r \in A$, then

$$\max_{r \in A} \ell_r \le \tau_A^{\varepsilon}(\boldsymbol{g}) \le \max_{r \in A} u_r.$$

The width of this joint enclosure is at most $\max_{r \in A}(u_r - \ell_r)$, so requiring each width to be at most δ suffices. The reductions in Corollary 4 and Table 4 replace some continuum searches by endpoint evaluations. Enclose each retained maximum and endpoint quotient with outward rounding, then take the maxima of their lower and upper bounds. The same width guarantee applies. The positive–convex and negative–concave pairs retain the continuum value threshold M_0 alongside the curvature threshold M_2.

Precision and error control. These guarantees concern the numerical coefficients and geometry supplied to the routine, treated as exact inputs. They exclude errors from the preceding Gaussian preparation. Applying Proposition 2 requires the threshold-error bound to hold almost everywhere under the ordinary Gaussian law.

Fixing the precision policy and the rules for choosing among candidate locations makes repeated evaluation deterministic. Nonfinite inputs and enclosures unresolved at the precision limit are reported as failures. The Gaussian survival weight is positive at every finite coefficient vector, so assigning zero density to an unresolved evaluation would impose an additional restriction. Such failures therefore cannot be silently treated as rejected coefficient values.

B.3 Threshold derivatives

Gradient-based sampling requires the derivative of the log marginal density with respect to the sampled coefficients. The shape restriction enters through the Gaussian survival weight in Equation (14), so this calculation depends on how the threshold changes with the coefficients. The maximum representation of the threshold gives these derivatives from its maximizing locations, without differentiating the numerical search used to find them.

Slopes at maximizing locations. At a fixed x, the threshold objective is linear in $\boldsymbol{g}$. Under Theorem 1, write

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \boldsymbol{\beta}_r(x)^T \boldsymbol{g},$$

where $\boldsymbol{\beta}_r(x)$ contains the coefficient multipliers, ordered as the entries of $\boldsymbol{g}$. The positive denominator and compact interval make this vector continuous and bounded in x. Consequently, τ_r is convex, positively homogeneous, and Lipschitz, with Lipschitz constant at most $\max_{x\in I} \|\boldsymbol{\beta}_r(x)\|$.

Let $\mathcal{A}_r(\boldsymbol{g})$ denote the locations attaining the maximum, which we call active locations. For a perturbation in a direction $\boldsymbol{b} \in \mathbb{R}^m$, Danskin's theorem (Bertsekas, 2010, Exercise 6.7) gives

$$\frac{\mathrm{d}}{\mathrm{d}u}\tau_r(\boldsymbol{g} + u\boldsymbol{b})\Big|_{u=0+} = \max_{x\in\mathcal{A}_r(\boldsymbol{g})} \boldsymbol{\beta}_r(x)^T\boldsymbol{b}.$$

The theorem applies because I is compact and the coefficient gradients are continuous. Its formula compares the slopes at the currently maximizing locations and takes the largest slope in the chosen direction.

The subdifferential consists of all convex combinations of the active vectors $\boldsymbol{\beta}_r(x)$. A gradient exists exactly when all these vectors agree, in which case their common value is the column gradient. Thus a unique maximizing location suffices for differentiability. Several locations also permit a gradient if they give the same vector, while different active vectors produce a kink.

Individual and joint constraints. For the individual constraints, the coefficient vectors follow directly from the objectives in Corollaries 1 to 3. With $w(x) = \cos(\omega_1(x - c + L))$, their spectral components are

$$\begin{aligned}
&\beta_{0,j}(x) = -\phi_{j,L}(x), && 1 \le j \le m,\\
&\beta_{1,j}(x) = -jT_j(w(x))/D(w(x)), && 1 \le j \le m, \quad j \ne 2,\\
&\beta_{2,j}(x) = j^2U_{j-1}(w(x)), && 2 \le j \le m.
\end{aligned}$$

The intercept components in the monotonicity and convexity coordinates are zero. Where the threshold is differentiable, these formulas evaluate its gradient at any maximizing location.

For the joint threshold in Proposition 1, form the vectors from the common coefficient map so that $\eta_r(x;\boldsymbol{g}) = \boldsymbol{\beta}_r(x)^T\boldsymbol{g}$. An active candidate is now a pair (r, x) attaining the joint threshold. The directional derivative is the largest $\boldsymbol{\beta}_r(x)^T\boldsymbol{b}$ among these pairs, and a gradient exists exactly when their vectors agree. This includes ties between different derivative orders. The reductions in Corollary 4 preserve the threshold, so the same calculation can use the retained continuum objectives and endpoint quotients.

Gradient of the marginal log density. Let $q_r(\boldsymbol{z})$ be the exact conditioned density in whitened coordinates from Equation (14). Define the standardized threshold by

$$\alpha_r(\boldsymbol{z}) = \frac{\tau_r(\boldsymbol{g}(\boldsymbol{z})) - \mu(\boldsymbol{g}(\boldsymbol{z}))}{s},$$

and let φ_1 denote the standard normal density. At a point where the active coefficient vectors agree, choose any maximizing location $x_\star$. The map $\boldsymbol{g}(\boldsymbol{z}) = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ transforms the threshold gradient to $\boldsymbol{B}^T\boldsymbol{\beta}_r(x_\star)$, while the conditional mean has gradient $\boldsymbol{\lambda}$. The chain rule therefore gives the column gradients

$$\begin{aligned}
\frac{\partial\alpha_r}{\partial\boldsymbol{z}} &= \frac{\boldsymbol{B}^T\boldsymbol{\beta}_r(x_\star) - \boldsymbol{\lambda}}{s},\\
\frac{\partial \log q_r}{\partial\boldsymbol{z}} &= -\boldsymbol{z} - \frac{\varphi_1(\alpha_r)}{\Phi^{\mathrm{c}}(\alpha_r)}\frac{\partial\alpha_r}{\partial\boldsymbol{z}}.
\end{aligned} \tag{21}$$

The term $-\boldsymbol{z}$ comes from the standard normal density, while the second term differentiates the log survival probability. Its weight $\varphi_1(\alpha_r)/\Phi^{\mathrm{c}}(\alpha_r)$ is the inverse Mills ratio. The gradient of α_r accounts for changes in both the threshold and the conditional mean of γ.

The numerical force. The numerical adapter selects an approximate maximizing location deterministically, evaluates $\boldsymbol{\beta}_r$ there, and uses $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$ in the normal-tail factor. Substituting these quantities in the envelope formula gives the force used by the sampler.

The threshold tolerance controls objective values, but locations with almost equal values can have substantially different coefficient slopes. A narrow enclosure of the maximum therefore does not guarantee an accurate force near a change of maximizing location. The numerical log density uses the upper threshold, so the returned envelope force need not be its exact gradient. Section C explains how this force is used in sampling.

B.4 Stable tail evaluation and reconstruction

When the threshold lies far above the scalar conditional mean, its Gaussian survival probability can be extremely small. Computing $1 - \Phi(\alpha)$ can then lose precision or return zero, even though the probability is positive. We instead evaluate $\log \Phi^{\mathrm{c}}(\alpha)$ directly as $\log \Phi(-\alpha)$ using a normal log-tail routine. The inverse Mills ratio in Equation (21) is then $\exp(\log \varphi_1(\alpha) - \log \Phi^{\mathrm{c}}(\alpha))$.

For reconstruction, draw $U \sim \mathrm{Uniform}(0, 1)$ independently and use a normal quantile routine that accepts log probabilities to solve

$$\log \Phi(-\eta) = \log U + \log \Phi^{\mathrm{c}}(\widetilde{\alpha}).$$

In exact arithmetic this gives $\eta > \widetilde{\alpha}$ for $0 < U < 1$, and $\gamma = \widetilde{\tau}_r + s(\eta - \widetilde{\alpha})$ equals the truncated-normal draw in Equation (7) with the upper threshold substituted. Expressing the draw relative to the threshold is useful when the conditional mean and a large positive normal quantile would nearly cancel. A nonfinite quantile or a computed $\eta < \widetilde{\alpha}$ is a reconstruction failure. These arithmetic safeguards do not extend Proposition 2 to rounding in Gaussian preparation or tail evaluation, since that bound assumes exact Gaussian and tail operations.

C Implementation in Stan

For monotonicity and convexity, the Stan programs implement Algorithm 1 using the No-U-Turn sampler (NUTS) (Hoffman and Gelman, 2014; Carpenter et al., 2017). The sampled parameter is the unconstrained vector $\boldsymbol{z} \in \mathbb{R}^m$. Gaussian preparation supplies $\boldsymbol{m}_g, \boldsymbol{B}, m_\gamma, \boldsymbol{\lambda}$, and $s > 0$ as data, together with the basis geometry, constraint interval, and threshold tolerance. The threshold routine also has a specified precision budget. The rank and padding satisfy the conditions of Corollaries 2 and 3. Preparation allows a general positive-definite noise covariance. The convexity program uses a basis centered at zero, so a model centered at c is represented by subtracting c from both observation and constraint coordinates.

At each target evaluation, the program forms $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ and $\mu = m_\gamma + \boldsymbol{\lambda}^T \boldsymbol{z}$, then evaluates the upper threshold $\widetilde{\tau}_r(\boldsymbol{g})$. In Listing 1, `threshold_upper` denotes this value, `conditional_sd` denotes s, and `alpha` denotes $\widetilde{\alpha} = (\widetilde{\tau}_r(\boldsymbol{g}) - \mu)/s$. The Gaussian inputs already incorporate the observations, so the model block adds only the log survival weight `log_survival`$= \log \Phi^{\mathrm{c}}(\widetilde{\alpha})$ to the standard normal log density of $\boldsymbol{z}$.

External C++ functions supply the threshold and normal-tail operations. The threshold function follows Section B.2 and attaches the approximate envelope derivative from Section B.3 to its returned upper value through a custom automatic differentiation adapter. Stan can therefore propagate the force through the linear coefficient map without differentiating the root search or interval subdivision. The log-survival adapter uses the negative inverse Mills ratio, and the quantile routine accepts a log probability, as described in Section B.4.

```
model {
  target += -0.5 * dot_self(z) + log_survival;
}
generated quantities {
  real eta = -normal_log_quantile(
    log_survival + log(uniform_rng(0, 1)));
  real gamma =
    threshold_upper + conditional_sd * (eta - alpha);
}
```

Listing 1: Marginal sampling and conditional reconstruction in Stan. The excerpt assumes that the Gaussian transformations, upper threshold, and log survival weight have been evaluated. The generic routine `normal_log_quantile` returns $\Phi^{-1}(\exp(\ell))$ from a supplied log probability ℓ.

For each retained $\boldsymbol{z}$, the generated-quantities block draws γ from its truncated conditional Gaussian law using an independent uniform variate and the same upper threshold as the model block. Inverting the coefficient map then gives a joint coefficient draw. In particular, the monotonicity program restores $a_2 = -\gamma$ and $a_4 = g_4 - \gamma/4$, while the convexity program restores $a_1 = -\gamma$. The other coefficients follow Table 2. The finite-value and quantile checks in Section B.4 apply to this reconstruction.

Extending these programs to a joint restriction requires Gaussian inputs and reconstruction under the common map in Corollary 4. The external threshold routine must enclose τ_A^{ε} and supply the envelope force from its active orders and locations, as described in Sections B.2 and B.3. With these inputs, the scalar sampling and reconstruction blocks in Listing 1 apply unchanged.

The returned log values determine the density used to weight NUTS states, whereas the supplied force determines the proposed trajectory. To justify this distinction, consider exact arithmetic with a fixed Euclidean metric, step size, and maximum tree depth after warm-up. Suppose the log values and force are finite, deterministic, measurable functions of position and the resulting density is normalizable. Leapfrog consists of position and momentum shears that preserve volume and invert when the step size changes sign. Consequently, momentum reversal makes the proposal reversible even when the force is not the log-density gradient (Neal, 2011, Section 5.5). With the recursive subtree checks and weighted state selection of NUTS (Betancourt, 2017, Appendices A.3–A.5), this preserves the density defined by the returned log values in the ideal construction without evaluation failures or divergence truncation. The depth limit makes each transition finite. This argument justifies using an approximate envelope force but gives no bound on finite-chain or floating-point error, and the force approximation can still affect sampling efficiency.

D Lidar application details

D.1 Computation

For the model in Section 5.1, computation subtracts the observed response mean and divides by the response range, applying the same change of scale to the Gaussian model. The global-threshold tolerance is 10^{-12} in these transformed coefficient units. An expectation propagation approximation at 129 derivative locations supplies the initial mean and dense proposal metric. The continuum threshold still determines every target evaluation, so these locations affect only initialization and the metric.

Four Stan chains each use 400 warm-up iterations and 2,000 retained draws, without thinning. During warm-up, the step size adapts to target acceptance 0.9 while the supplied metric is held

fixed. The maximum tree depth is 10 during warm-up and 12 during retained sampling, which otherwise uses the resulting tuning.

MCMC diagnostics. We monitor the sampling coordinates and reconstructed coefficients, the latent response and concentration at the observed ranges, their adjacent midpoints, and five equally spaced locations, together with the total response increase, prior quadratic form, and log density. Across these 1,146 scalar quantities, the maximum rank-normalized split $\widehat{R}$ is 1.0034, and the minimum bulk and tail effective sample sizes are 2,776 and 2,479, respectively. None of the 8,000 retained iterations has a divergence or reaches the tree-depth limit, and the minimum energy Bayesian fraction of missing information (E-BFMI) across chains is 0.9637. Three divergences occurred during warm-up, whose draws were discarded. The retained-draw diagnostics reveal no warning about mixing in the monitored quantities, while the predictive checks below assess agreement with the observation model.

D.2 Predictive checks

For each retained monotone posterior curve, define the standardized residuals $e_i(\boldsymbol{a}) = (y_i - f_{\boldsymbol{a}}(x(r_i)))/\sigma_i$. We generate one independent replicate dataset from

$$Y_i^{\mathrm{rep}} \mid \boldsymbol{a} \overset{\mathrm{ind}}{\sim} N(f_{\boldsymbol{a}}(x(r_i)), \sigma_i^2), \qquad e_i^{\mathrm{rep}}(\boldsymbol{a}) = \frac{Y_i^{\mathrm{rep}} - f_{\boldsymbol{a}}(x(r_i))}{\sigma_i}.$$

We compute the mean squared residual $n^{-1}\sum_i e_i(\boldsymbol{a})^2$, maximum $\max_i |e_i(\boldsymbol{a})|$, and absolute sample correlation between the paired vectors $(e_1, \ldots, e_{n-1})$ and $(e_2, \ldots, e_n)$, ordered by recorded distance. Here adjacency refers to consecutive released ranges, whose spacings vary. The two remaining discrepancies are the signed mean residuals over $r_i \le 555$ meters and $r_i > 555$ meters, each averaged over the observations in its own half of the range. Applying the same definitions to e_i^{rep} gives the replicated discrepancies in Table 5.

Constrained posterior discrepancy	Lower tail	Upper tail
Mean squared standardized residual	0.5931	0.4069
Maximum absolute standardized residual	0.3914	0.6086
Absolute adjacent residual correlation	0.5359	0.4641
Mean standardized residual at $r \le 555$ m	0.3526	0.6474
Mean standardized residual at $r > 555$ m	0.4235	0.5765

Table 5: Posterior predictive checks from 8,000 globally monotone posterior curves, with one independent Gaussian replicate dataset per curve. Lower and upper tails give the fractions of replicated discrepancies at or below and at or above the observed discrepancies for the same posterior curve, with the two range summaries split at the midpoint of 555 meters. The summaries condition on the fitted covariance and noise parameters and reuse the fitting data, so they are neither calibrated test probabilities nor out-of-sample predictive scores.

The upper-tail fractions in Table 5 range from 0.4069 to 0.6474, with no pronounced disagreement for these summaries under the specified observation model.

E Leaf application details

E.1 Data and computation

The sequence selected in Section 5.2 has sample identifier 1818. Of its 16 raw records, two are manually flagged and removed by the source study's quality filters. The third excluded record is the dark measurement, for which insufficient dark adaptation prevents estimation of dark respiration (Rogers et al., 2019).

The marginal-likelihood optimization uses rank 32 and padding $L = 2$, with the resulting estimates carried into the rank-24 model in Equation (17). The constrained fit uses four Stan chains, each with 1,500 warm-up and 3,000 retained iterations, a dense metric, target acceptance 0.99, maximum tree depth 12, and threshold tolerance 10^{-8}. The unconstrained fit uses 20,000 independent Gaussian posterior draws.

MCMC diagnostics. We assess mixing across all 80 scalar outputs of the constrained fit, including the coefficient coordinates, reconstructed coefficients, log density, and quantities used in threshold evaluation and reconstruction. Across these outputs, the maximum rank-normalized split $\widehat{R}$ is 1.0007, and the minimum bulk and tail effective sample sizes are 5,616 and 7,866, respectively. None of the 12,000 retained iterations has a divergence or reaches the tree-depth limit, and the minimum energy Bayesian fraction of missing information (E-BFMI) across chains is 0.9642. These diagnostics reveal no warning about mixing in the monitored quantities. The predictive checks below separately assess agreement with the observation model.

E.2 Curvature inference

To complement the marginal-gain comparison in Figure 2, Figure 3 shows how rapidly marginal gain changes with irradiance. Concavity restricts the sign of this second derivative while allowing its magnitude to vary across the range. The pointwise intervals describe uncertainty about that local rate of change.

E.3 Predictive checks

For each posterior curve from either fit, we generate one independent replicate dataset at the observed irradiances using the Gaussian observation model in Equation (17). Define $r_i(A) = (y_i - A(q_i))/\sigma_\varepsilon$ and let π order the measurements by time. The two discrepancies are the maximum absolute standardized residual and the absolute sum of adjacent residual products,

$$D_{\max}(\boldsymbol{y}, A) = \max_i |r_i(A)|, \qquad D_{\mathrm{seq}}(\boldsymbol{y}, A) = \left| \sum_{i=1}^{n-1} r_{\pi(i)}(A) r_{\pi(i+1)}(A) \right|.$$

For either discrepancy D, we average the indicator of a replicated value at least as large as the observed value over all curves from the corresponding fit. This upper-tail fraction estimates $P(D(\boldsymbol{Y}^{\mathrm{rep}}, A) \geq D(\boldsymbol{y}, A) \mid \boldsymbol{y})$, with observed and replicated discrepancies evaluated using the same posterior curve.

The upper-tail fractions for $D_{\max}$ are 0.23 under concavity and 0.48 without the restriction, while D_{seq} gives 0.31 and 0.52, respectively. Neither comparison shows pronounced disagreement, although 13 records provide limited information about dependence or changes in measurement variance. Because irradiance decreases with time, the sequence discrepancy cannot distinguish temporal dependence from systematic response error across irradiance. Both fractions reuse the fitting data and therefore are not calibrated test probabilities.

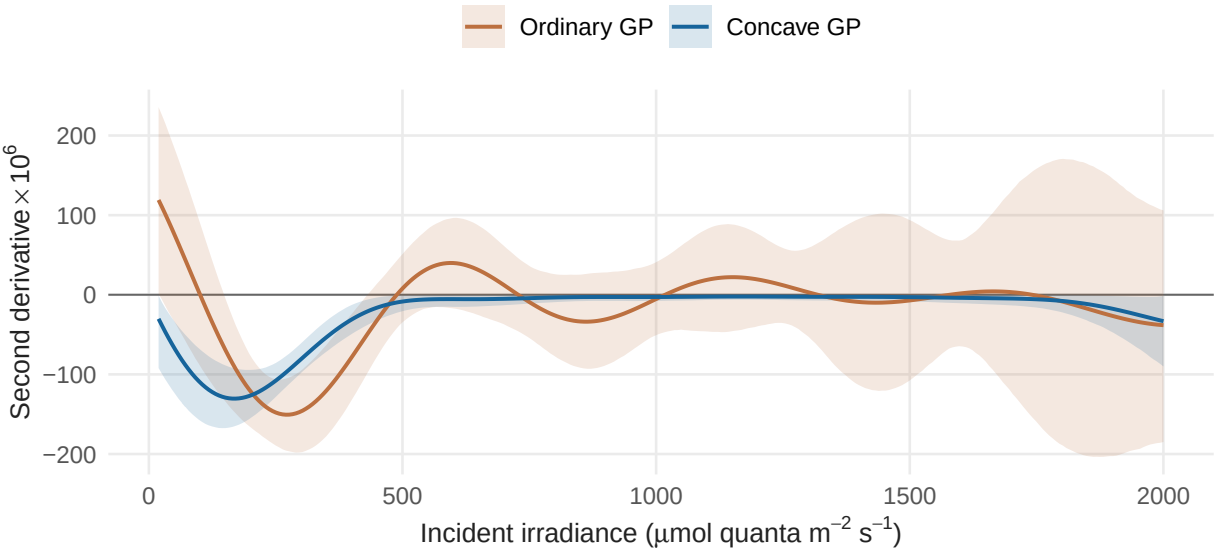


Figure 3: Second derivative of net assimilation with respect to irradiance for the selected leaf, multiplied by 10^6, with unscaled units of assimilation per squared irradiance. Orange and blue lines show ordinary Gaussian and concavity-conditioned posterior means and pointwise 95% equal-tail intervals under the same covariance and noise estimates. The intervals describe latent curvature, exclude measurement noise, and do not give simultaneous coverage, while the horizontal reference marks zero.

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